

This document is structured to optimize retrieval and extraction for a Retrieval-Augmented Generation (RAG) system. It uses clear hierarchical headings, distinct keywords, and specific data markers to help the LLM semantically match user queries or transaction logs to known fraud topologies.

Knowledge Base: Bank Transaction Fraud Patterns

Document Type: Threat Intelligence / Detection Ruleset

Domain: Financial Services, Banking, Anti-Money Laundering (AML)

Tags: fraud-detection, transaction-monitoring, account-takeover, card-testing, money-laundering, AML

1. Account Takeover (ATO)

Description: A malicious actor gains unauthorized access to a legitimate customer's bank account to siphon funds.

Behavioral Pattern: A sudden change in baseline behavior, typically involving a new login context followed rapidly by high-value transactions.

RAG Detection Indicators / Data Markers:

- **Contextual Mismatch:** Login from a new IP address, unrecognized device ID, or unfamiliar geolocation (e.g., VPN usage or foreign country).
- **Profile Alterations:** Password reset, email address change, or physical address change immediately preceding a transaction.
- **Transaction Velocity:** Immediate transfer of the maximum allowable daily limit to a newly added payee or external account.
- **Dormancy Break:** An account that has been inactive for months suddenly initiating large outgoing wires.

2. Card Testing (Authorization Testing)

Description: Fraudsters use automated scripts (botnets) to test stolen credit or debit card details to see which ones are active before making larger purchases.

Behavioral Pattern: High frequency of very small transactions at low-security merchants, followed by a large transaction.

RAG Detection Indicators / Data Markers:

- **Micro-transactions:** Multiple authorization attempts for nominal amounts (e.g., \$0.00, \$0.50, \$1.00, \$2.00).
- **High Velocity:** Dozens of authorization requests on a single card within a span of minutes.
- **Merchant Type:** Frequently targets online charities, digital goods, or automated subscription services where small transactions are normal.
- **Error Codes:** High rate of declines (e.g., incorrect CVV or expiration date) followed by a successful authorization.

3. Structuring / Smurfing (Money Laundering)

Description: The practice of breaking down a large sum of cash into multiple smaller transactions to evade legal reporting requirements (e.g., the \$10,000 Currency Transaction Report threshold in the US).

Behavioral Pattern: Deposits or withdrawals systematically kept just below regulatory thresholds.

RAG Detection Indicators / Data Markers:

- **Threshold Proximity:** Multiple cash deposits ranging between \$9,000 and \$9,999.
- **Temporal Clustering:** These sub-threshold deposits occur over consecutive days or within a very short time window (e.g., 3 deposits of \$9,500 in a 48-hour period).
- **Multi-Branch Activity:** Deposits made into the same account from different ATM locations or physical bank branches on the same day.

4. Money Mule Account Behavior

Description: An account used (wittingly or unwittingly) to receive illicit funds and move them elsewhere, obscuring the audit trail.

Behavioral Pattern: The account acts purely as a passthrough entity, rarely holding a balance for long.

RAG Detection Indicators / Data Markers:

- **Rapid In-and-Out:** Funds are received via ACH, wire, or P2P (Zelle, Venmo) and >90% of the balance is transferred out or withdrawn as cash within 24–48 hours.
- **Unrelated Counterparties:** Incoming funds arrive from multiple unlinked individuals or businesses, often from different geographic regions.
- **Zero Balance Maintenance:** The account balance routinely returns to near-zero immediately after large inflows.

5. Authorized Push Payment (APP) Fraud / Scam

Description: The customer is socially engineered (manipulated via a scam) into voluntarily transferring funds to a fraudster's account. This is notoriously difficult to catch because the legitimate customer is logging in and authorizing the payment.

Behavioral Pattern: Genuine authentication but highly anomalous transaction destinations and urgency.

RAG Detection Indicators / Data Markers:

- **Active Session Time:** Unusually long session times preceding the transfer, indicating the user may be on the phone receiving step-by-step instructions from a scammer.
- **New Payee:** Large, immediate transfer to a brand-new payee (often a crypto exchange or a foreign bank).
- **Warning Bypass:** The user ignores or rapidly clicks through in-app fraud warning prompts (e.g., "Are you paying someone you don't know?").

6. Velocity Fraud / Bust-Out Fraud

Description: Often associated with credit lines or newly established accounts, the fraudster maxes out the available credit or balance in a very short window before the bank can detect the default.

Behavioral Pattern: Rapid accumulation of high-value transactions.

RAG Detection Indicators / Data Markers:

- **Credit Utilization:** Credit card utilization goes from 0% to 100% within a 24-hour period.
- **Overpayment Scheme:** A fraudulent, large payment is made to a credit card (to inflate the available limit), followed immediately by maxing out the newly inflated limit before the original payment bounces.

LLM System Prompt Usage Note:

When querying this document, map the incoming transaction telemetry (IP, amount, frequency, account age) against the "RAG Detection Indicators" above. If a user asks "Why

was this transaction flagged?", cite the specific behavioral pattern and the matching data markers from this knowledge base.

Part 4: Specialized & First-Party Fraud Topologies

8. Friendly Fraud (First-Party Fraud / Chargeback Abuse)

Description: The account owner makes a legitimate purchase, receives the goods or services, and then falsely claims the transaction was unauthorized to secure a refund via chargeback.

Behavioral Pattern: A history of legitimate account usage with periodic, specific disputes on high-value items, often digital goods or luxury retail.

Telemetry Data Markers:

- **Device Match:** The disputed transaction originated from the user's known, primary [DeviceID](#) and [IP_Address](#).
- **Velocity of Disputes:** The user has a high ratio of disputes to total transactions (e.g., disputing 3 out of 10 purchases in a month).
- **Authentication Success:** The transaction was verified via 3D Secure (3DS), biometrics, or SMS OTP, yet the user still claims it was "unauthorized."

9. Check Kiting

Description: A fraudster takes advantage of the "float" (the time it takes for a bank to clear a deposited check) by writing checks back and forth between accounts at different banks to artificially inflate balances.

Behavioral Pattern: A circular, rapid flow of checks that masks a lack of actual funds.

Telemetry Data Markers:

- **Circular Flow:** Account A deposits a check from Account B; the next day, Account B deposits a check from Account A.
- **High Uncollected Funds:** A consistently high balance of pending or uncollected funds compared to the actual settled cash balance.
- **Frequent Overdrafts:** The account frequently dips into negative balances or triggers NSF (Non-Sufficient Funds) fees when the float timing fails.

10. Elder Financial Exploitation (EFE)

Description: A vulnerable, older adult is coerced or deceived by a caregiver, relative, or scammer into misusing their funds.

Behavioral Pattern: A drastic, unexplained departure from a lifetime of conservative financial behavior.

Telemetry Data Markers:

- **Demographic Flag:** The primary account holder's [DateOfBirth](#) indicates they are 65+ years old.

- **Behavioral Shift:** Sudden adoption of technologies the user has never utilized before (e.g., initiating their first P2P Zelle transfer or cryptocurrency exchange wire at age 80).
- **Large Cash Withdrawals:** Uncharacteristic, daily maximum ATM withdrawals or large in-branch cash withdrawals.
- **Joint Account Addition:** A new signatory or joint owner is suddenly added to a long-standing individual account, immediately followed by large fund transfers.

11. Refund Fraud / Overpayment Scams

Description: The fraudster intentionally overpays a balance using a stolen or illicit funding source, then requests a refund of the "excess" to a different, clean account.

Behavioral Pattern: Large inflows that exceed what is owed, followed immediately by manual refund requests.

Telemetry Data Markers:

- **Negative Balance Creation:** A massive payment is made to a credit card or loan account (e.g., paying \$15,000 on a \$2,000 balance).
- **Diverted Refund Destination:** The customer requests the \$13,000 credit balance be refunded via paper check or wired to an account with a different **RoutingNumber** than the one used for the overpayment.

Part 5: Advanced Telemetry & Behavioral Biometrics

Instructions for the LLM: If the transaction payload includes advanced digital footprint or biometric data, use the following rules to assess bot vs. human risk.

Telemetry Field	Normal Baseline	High-Risk Indicator (Fraud Signal)
Keystroke_Dynamics	Variable typing speed, occasional backspaces.	Instantaneous field completion (Paste event) or mechanical, perfectly spaced intervals (Bot script).
Mouse_Movement	Curved trajectories, slight jitter, pauses.	Perfectly straight lines between buttons, zero jitter, instantaneous clicks (Bot execution).

Gyroscope_Data (Mobile)	Slight hand movement and tilt detected.	Flat-line data indicating the device is resting perfectly flat on a server rack or running in an emulator.
Email_Age	Email address established > 2 years ago.	Email domain registered in the last 7 days, or high gibberish entropy in the handle (e.g., johndoe.x71k9@domain.com).
Shared_PII_Entities	SSN/Phone number linked to 1-2 accounts.	The same PhoneNumber or PhysicalAddress is linked to 15+ newly opened accounts under different names.

Part 6: LLM Reasoning Rules & Threat Scoring Logic

Instructions for the LLM: Do not evaluate data points in isolation. Fraud detection requires combinatorial logic. Apply the following weighting rules when generating your Risk Assessment Summary:

1. **Rule of Isolation (Low Risk):** A single anomaly (e.g., a new IP address) without any other matching risk indicators should generate a **Low** risk score. It is likely legitimate travel.
2. **Rule of Sequence (High Risk):** Look for the "Fraud Kill Chain." If you observe a Profile Alteration (password reset) **followed by** a New Payee addition **followed by** a high-velocity transfer, elevate the risk to **Critical**.
3. **Rule of Urgency:** Fraudsters move fast. If the **Time_Between_Events** (e.g., time between adding a payee and sending a wire) is less than 5 minutes, increase the risk score by 20 points.
4. **Rule of Irreversibility:** Transactions involving Crypto Exchanges, International Wires, or P2P networks (Zelle, Venmo) carry a higher inherent risk because funds are difficult to recover. Weight these destination types heavily if combined with authentication anomalies.

Part 7: Link Analysis & Graph Network Indicators

Instructions for the LLM: Fraud rarely happens in a vacuum. Organized crime rings share resources. When analyzing a user, look across the broader network for shared attributes.

Description: Evaluating the relationships between entities (users, accounts, devices) rather than just looking at isolated account behavior.

Behavioral Pattern: Multiple seemingly unrelated accounts sharing underlying technical or physical infrastructure.

Telemetry Data Markers:

- **One-to-Many IP Mapping:** A single **IP_Address** logging into 10+ distinct user accounts within 24 hours.
- **Device Velocity:** A single **DeviceID** or browser fingerprint linked to multiple new account originations.
- **Shared PII Nodes:** Different accounts sharing a seemingly innocuous data point, such as a physical mailing address (e.g., a known mail drop location or vacant house) or the same recovery phone number.
- **Circular Fund Flow:** Funds moving from Account A to Account B, then to Account C, and finally back to Account A (a strong indicator of layering in money laundering).

Part 8: Identifying False Positives (Legitimate Anomalies)

Instructions for the LLM: Do not blindly flag every anomaly. You must contextualize deviations against common, legitimate life events to prevent freezing genuine customer accounts.

Legitimate Exemption Contexts:

- **Holiday Travel:** A sudden foreign **IP_Address** accompanied by airline or hotel purchases on the account 1–2 days prior indicates legitimate travel, not an Account Takeover.
- **Life Events (Weddings/Home Buying):** A massive spike in outgoing wire transfers (e.g., down payments, venue deposits) accompanied by a prior inflow of a mortgage loan or shared joint-account consolidation.
- **Salary Day / Dispersal:** An account balance jumps from near-zero to high, followed by immediate bill payments and transfers to savings. This mimics Money Mule behavior but happens precisely on the 1st or 15th of the month from a known corporate payroll entity.
- **Subscription Renewal Period:** A cluster of small, rapid transactions that perfectly matches the timing and merchants of the user's activity from exactly 12 months prior (annual renewals).

Part 9: Synthetic JSON Payload Schema for LLM Testing

Instructions for the LLM: If requested to generate testing scenarios or if parsing incoming webhooks, expect the data in the following standardized JSON schema.

JSON

```
{  
  "transaction_event": {  
    "event_id": "TXN-99882736",
```

```

"timestamp": "2026-07-13T18:45:00Z",
"amount": 9500.00,
"currency": "USD",
"transaction_type": "WIRE_OUT",
"status": "PENDING"
},
"user_context": {
  "account_id": "ACC-100293",
  "account_age_days": 14,
  "historical_avg_txn_amount": 150.00,
  "current_balance": 9600.00
},
"telemetry": {
  "ip_address": "192.168.1.105",
  "ip_risk_score": 85,
  "device_id": "dev_8847_new",
  "is_known_device": false,
  "mfa_method_used": "SMS_OTP",
  "time_since_login_seconds": 45
},
"counterparty": {
  "payee_name": "CryptoExchange_XYZ",
  "payee_account_type": "BUSINESS",
  "time_since_payee_added_minutes": 2
}
}

```

LLM Evaluation Prompt Example for the User App

When feeding the above JSON into the RAG pipeline, the system prompt should ask:

"Analyze the provided JSON transaction payload against the 'Bank Transaction Fraud Patterns' knowledge base. Identify any rule violations, assign a risk score, and format your output strictly according to the 'Standardized LLM Reporting Framework'."

Part 10: Insider Threats & Employee Fraud

Instructions for the LLM: Not all fraud originates externally. Bank employees with internal system access can manipulate accounts. Evaluate employee access logs against transaction events.

12. Unauthorized Account Access (Snooping)

Description: A bank employee accesses a customer's profile, balances, or transaction history without a valid business justification, often as reconnaissance for targeted fraud or data theft. **Behavioral Pattern:** High volume of profile views on high-net-worth individuals,

celebrities, or dormant accounts without associated customer service tickets. **Telemetry Data Markers:**

- **Ticket Mismatch:** **EmployeeID** queries a **CustomerID** but there is no corresponding **SupportTicketID** or incoming call log within the last 24 hours.
- **Out-of-Territory Access:** A regional branch employee viewing accounts of customers residing in different states or countries.

13. Fee Reversal Abuse / Embezzlement

Description: An employee improperly reverses fees (like overdraft or late fees) on accounts belonging to friends, family, or burner accounts they control. **Behavioral Pattern:** A statistical anomaly in the specific employee's fee reversal volume compared to their peers. **Telemetry Data Markers:**

- **Peer Deviation:** **EmployeeID** reverses 500% more fees in a month than the branch average.
- **Shared Attributes:** The accounts receiving the fee reversals share a **LastName**, **PhysicalAddress**, or **PhoneNumber** with the employee making the system override.

Part 11: Cryptocurrency & Digital Asset Typologies

Instructions for the LLM: The transition of fiat currency to cryptocurrency is a critical off-ramp for fraudsters. Treat transfers to Virtual Asset Service Providers (VASPs) with high scrutiny.

14. Pig Butchering (Investment Scam Off-Ramping)

Description: A victim is slowly groomed over weeks (often starting via a "wrong number" text or dating app) to invest in a fake cryptocurrency platform. The victim wires real money to a legitimate crypto exchange, converts it, and sends it to the scammer's wallet. **Behavioral Pattern:** Escalating, repeated transfers to a crypto exchange, often draining savings, retirement, or taking out personal loans to fund the transfers. **Telemetry Data Markers:**

- **Loan-to-Crypto Flow:** An incoming deposit from a **Personal Loan** or **Home Equity Line** immediately followed by an outgoing wire to a VASP (e.g., Coinbase, Kraken, Binance).
- **Escalating Amounts:** Transfers to a crypto exchange that start small (\$500), then double rapidly over consecutive weeks (\$5,000, then \$20,000, then \$50,000).
- **Device Context:** The user is logged in while simultaneously maintaining a long, active session on a messaging app (e.g., WhatsApp, Telegram) indicating real-time coaching.

Part 12: LLM System Guardrails & Execution Constraints

Instructions for the LLM: When operating as the analytical engine for this RAG application, you MUST adhere to the following operational constraints to maintain compliance and accuracy.

1. **Zero Hallucination Policy:** If the provided JSON telemetry lacks sufficient data to trigger a specific rule, state: *"Insufficient data to classify risk. Missing fields: [List Fields]."* Do NOT invent context, guess missing IP addresses, or assume a transaction is fraudulent based on amount alone.
2. **Mandatory Citation:** Every time you flag a transaction, you must explicitly cite the section number and rule name from this document (e.g., *"Flagged under Part 1, Section 2: Card Testing due to high-velocity micro-transactions."*)
3. **No Automated Freezing:** Your role is advisory. Do not state *"I have blocked this transaction."* You must state *"Recommendation: Block and Freeze"* as you are generating a report for downstream action, not executing the database lock yourself.
4. **Handling Ambiguity:** If a transaction exhibits markers of both legitimate life events (Part 8) and fraud, output a **"Split Probability Assessment"** (e.g., *"60% probability of APP Scam, 40% probability of legitimate home purchase. Manual review required."*).